

Rana Zohreh

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<https://github.com/ranazohreh>

RESEARCH INTERESTS

My research uses machine learning and computational transcriptomics to study gene regulation, grounded in wet-lab molecular biology. I build gene-regulatory and co-expression networks from RNA-seq and public multi-omics data, apply and interpret machine-learning models to connect regulatory programs to cellular phenotype, and have hands-on experimental training. I am eager to extend this work into single-cell and spatial transcriptomics, linking gene-regulatory programs to cell identity and function, and to keep combining computational modeling with experimental work.

EDUCATION

M.Sc. Molecular Genetics

2021–2025

Islamic Azad University, Central Tehran Branch, Tehran, Iran

GPA: 18.52 / 20

- Thesis: *Comprehensive Comparison of Common Mutations and Gene Networks in Prostate Cancer and Testicular Cancer Using Bioinformatics Tools (graded Excellent)*. Supervisor: Dr. Nastaran Asghari Moghaddam; Advisor: Dr. Mina Ramezani

B.Sc. Molecular Genetics

2016–2020

Islamic Azad University, Qom, Iran

GPA: 15.84 / 20

RESEARCH EXPERIENCE

Graduate Researcher — Population-Specific ceRNA Networks in Prostate Cancer

2024–present

Department of Biology, Central Tehran Branch, Islamic Azad University, Tehran, Iran

- Constructed population-specific gene-regulatory (ceRNA) networks from RNA-seq (GSE104131; DESeq2), integrating HMDD v4, starBase, and four target-prediction databases.
- Validated regulatory axes in TCGA-PRAD via Spearman/Pearson correlation with BH-FDR, expression-matched background pairs (Wilcoxon), and Fisher Z cross-population testing.
- Prioritized axes with a 4-criterion composite score (statistical robustness, miRTarBase, PPI hub/MCODE topology, CTD-curated disease association); identified druggable targets via custom DGIdb v5 GraphQL API integration.

Researcher — Machine Learning for Early-Onset Prostate Cancer Biomarkers

2024–2025

Department of Biology, Central Tehran Branch, Islamic Azad University, Tehran, Iran

- Integrated WGCNA co-expression analysis with DESeq2 differential expression on TCGA-PRAD (White patients, <55 yrs); identified 485 candidate genes at the WGCNA driver–DEG intersection, enriched in cytoskeleton-organization pathways.
- Built a nested leave-one-out cross-validation pipeline with LASSO feature selection across SVM, Random Forest, and ANN classifiers (AUC 0.86 / 0.89 / 0.91, respectively).
- Identified the lncRNA PCSEAT as the only biomarker common to all three classifiers in a White early-onset prostate cancer cohort; confirmed feature robustness with permutation importance and SHAP, and validated its tumour overexpression in an independent dataset (GSE115414).

MSc Thesis Researcher — Comparative Cancer Genomics

2021–2025

Department of Biology, Central Tehran Branch, Islamic Azad University, Tehran, Iran

- Compared African American vs. European American patients across prostate (TCGA-PRAD) and testicular (TCGA-TGCT) cohorts, integrating mRNA and lncRNA (plus miRNA for PRAD) expression with somatic mutation profiles (cBioPortal).
- Constructed PPI and ceRNA networks for all four population-cohort groups; identified population-specific hub genes and distinct pathways by population and tumor type.

Laboratory Intern — Wet-Lab Training

2023

GenIran, Tehran, Iran

- Hands-on training in molecular biology techniques (DNA/RNA extraction, PCR, qRT-PCR, gel electrophoresis) and mammalian cell culture (passaging, freezing/thawing, MTT and AO/PI viability assays).

INDEPENDENT PROJECT

- Spatial transcriptomics of mouse brain: scanpy/squidpy preprocessing, clustering, and spatial-domain analysis, with a PyTorch cell-type classifier benchmarked against a scikit-learn baseline and interpreted via feature importance. GitHub: github.com/ranazohreh/spatial-neuro-cell-identity

PUBLICATIONS

Peer-Reviewed Articles

Saeidian AH, Hooshmand A, Dehghan F, **Zohreh R**, Farahani A, Khamineh Y, Bagherabadi A, Rigobello R, Farhadieh ME, Dai H, Youssefian L, Vahidnezhad H, Akbarzade M, Hakonarson H, Zhao X. *From Aberrant Transcripts to Clinical Applications: The Era of RNA-Seq in Rare Disease Diagnosis*. *Clinical Chemistry*, accepted, 2026.

Zohreh R, Rezaei N, Amirifar P. *DNA Methylation in Glioblastoma: Insights from Single-Cell Epigenomics*. *Epigenomics*, accepted, 2026.

Manuscripts Under Review

Meskar Z, Yousefi MJ, **Zohreh R***, Jabbaripour Sarmadian A, Yazdanpanah N, Saleki K, Shams-Nateri A, Rezaei N. *The Multifaceted Role of NEK7: Mitosis, Inflammation, and Cancer Immunopathology*. *Cancer Cell International*, under review, 2025. *Co-second author; equal contribution.

Manuscripts in Preparation

Zohreh R, Asghari Moghaddam N, Ramezani M. *Population-Stratified ceRNA Network Analysis Reveals Distinct Regulatory Axes and Druggable Targets in African American and European American Prostate Cancer*. In preparation.

CONFERENCE PRESENTATIONS

Zohreh R, Asghari Moghaddam N, Ramezani M. *Precision diagnosis in early-onset prostate cancer: a machine learning and transcriptomic approach identifies PCSEAT lncRNA as a race-specific biomarker*. E-Poster (EP17.007), European Human Genetics Conference (ESHG 2025), Milan, Italy, 24–27 May 2025.

Zohreh R, Asghari Moghaddam N, Ramezani M. *Identifying hsa-miR-17-3p as a Potential Biomarker in Prostate Cancer Patients' Body Fluids Using Bioinformatics Tools*. 6th International & 18th Iranian Genetics Congress, Tehran, Iran, 2024.

Zohreh R, Amanzadeh Jajin E. *Decoding Glioblastoma Treatment Responses: A Combined Bioinformatics and Machine Learning Study*. Oral presentation, 2nd International Congress of Cancer Genomics (CGC2024), Tehran, Iran, 23–25 October 2024.

Zohreh R, Ayareh N, Montazeri Moghaddam F, Amanzadeh Jajin E. *Differentiation of Glioblastoma Multiforme Solid and Recurrent Tumors Using Gene Expression Profile and Machine Learning Algorithms*. 3rd International & 12th National Iranian Conference on Bioinformatics, 28–29 February 2024.

Montazeri Moghaddam F, **Zohreh R**, Amanzadeh Jajin E. *Prediction of Prognosis in Glioblastoma Patients Treated with Chemotherapy and Radiotherapy Using Genomic SNV Profile*. 6th International & 18th Iranian Genetics Congress, Tehran, Iran, 2024.

HONORS & AWARDS

Best Oral Presentation Award

October 2024

2nd International Congress of Cancer Genomics (CGC2024), Tehran, Iran. Awarded for *Decoding Glioblastoma Treatment Responses: A Combined Bioinformatics and Machine Learning Study*.

COMPUTATIONAL SKILLS

Programming: R, Python

Transcriptomics & RNA-seq: DESeq2, edgeR, limma, TCGAbiolinks, WGCNA, GSEA, clusterProfiler, variance-stabilizing normalization, batch-effect correction

Network & Functional Analysis: gene-regulatory / ceRNA networks, Spearman/Pearson correlation, Fisher Z-test, GO/KEGG enrichment, PPI hub analysis (Cytoscape, STRING, GeneMANIA, CytoHubba, MCODE)

Machine Learning: LASSO, SVM, Random Forest, XGBoost, ANN; nested leave-one-out cross-validation; permutation importance, SHAP; scikit-learn, Single-cell/spatial (scanpy, squidpy) and deep-learning (PyTorch) workflows.

Survival Analysis: Cox proportional hazards, time-dependent ROC/AUC (R survival package)

Drug-Gene Interaction: DGIdb v5 via GraphQL API (custom integration); CTD database (MeSH-based curation)

Visualization: ggplot2, matplotlib, volcano/MA plots, heatmaps, PCA /UMAP, GraphPad Prism

LABORATORY SKILLS

Nucleic acid extraction (phenol-chloroform, salting-out, kit-based for DNA; TRIzol and kit-based for RNA); cDNA synthesis; conventional, RFLP, ARMS, Tetra-ARMS, and Real-Time PCR (qRT-PCR); primer design (GeneRunner); agarose and PAGE gel electrophoresis with silver staining; mammalian cell culture (passaging, freezing, thawing); MTT and Acridine Orange/Propidium Iodide viability assays.

LANGUAGES

Persian: Native **English:** Professional working proficiency